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STD 8 - BASIC SCIENCE - UNIT 3: CELLS AND CELL CLUSTERS

(Model Question Paper)

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Total Marks: 20

SECTION A

(Questions 1 and 2 carry 1 mark each) (2 x 1 = 2)

1. Analyse the statements i) to iv) related to the History of Cell Biology and choose the correct answer from the options given below.

i) Robert Hooke first observed 'cells' in a thin slice of cork.

ii) Matthias Jakob Schleiden introduced the concept that new cells are formed from pre-existing cells.

iii) Anton van Leeuwenhoek discovered micro-organisms in pond water using a better microscope.

iv) The Cell Theory states that the cell is the basic unit of life.

a. i, ii incorrect, iii, iv correct

b. i, iii, iv correct, ii incorrect

c. ii, iii, iv correct, i incorrect

d. i, ii, iii, iv correct

2. Choose and write the correct answer from the options given below where the pairs are correctly matched.

I. Cell Organelle	**II. Function**
P. Endoplasmic Reticulum	i) Stores water, nutrients and wastes
Q. Mitochondria	ii) Center for protein synthesis
R. Vacuole	iii) Energy production centre of the cell
S. Ribosome	iv) Pathway for conducting materials

	(P)	**(Q)**	**(R)**	**(S)**
a.	(iv)	(iii)	(i)	(ii)
b.	(iii)	(iv)	(ii)	(i)
c.	(ii)	(i)	(iii)	(iv)
d.	(iv)	(iii)	(ii)	(i)

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SECTION B

(Questions 3 to 8 carry 2 marks each) (6 X 2 = 12)

3. Analyse the given instruments and answer the questions.

P: Compound Microscope, Q: Electron Microscope

i) Which instrument has the ability to magnify objects more than a million times? (1 mark)

ii) Name the type of lens used in instrument P and the type of lens used in instrument Q ? (1 mark)

4. A plant cell is enclosed by two protective layers. One is a rigid layer and the other is a semi-permeable layer.

i) Name the rigid layer outer to the cell membrane and state its primary function. (1 mark)

ii) Why is the flexible layer (Cell membrane) known as a semi-permeable membrane? (1 mark)

5. Plant tissues are broadly classified into simple and complex tissues.

i) Define a simple tissue. (1 mark)

ii) Why are Xylem and Phloem called complex tissues? (1 mark)

6. Analyse the following statements about cell organelles and fill in the blanks.

i) The cell organelle that appears as a network of threads within the nucleoplasm is the ..... which condenses into ..... during cell division. (1 mark)

ii) State the specific function of the Golgi Apparatus regarding the transport of proteins and lipids. (1 mark)

7. Animal tissues are essential for performing various functions in the body.

i) Which tissue is composed of cells capable of contraction and relaxation and helps in body movement? (1 mark)

ii) State the functions of Neural tissue. (1 mark)

8. Analyse the indicators and write answers to the questions.

Indicators:

- Cells with abundant cytoplasm.
- Large nucleus.
- Thin cell wall.

  i) Which type of plant tissue has cells with these characteristics? (1 mark)

  ii) State the main function of this tissue. (1 mark)

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SECTION C

(Questions 9 and 10 carry 3 marks each) (2 X 3 = 6)

9. Stem cells are often referred to as the body's 'master cells'.

i) Why are stem cells called the body's master cells? (1 mark)

ii) State any two ways in which stem cells help maintain healthy tissues in the body. (2 marks)

10. (A). Plant tissues are classified into Parenchyma, Collenchyma, and Sclerenchyma based on their cell structure.

Compare these three simple tissues based on the following indicators (3 marks):

- Cell type (Living/Dead)
- Cell wall nature
- Function

OR

10. (B). Plastids are specific parts in plant cells that help in the synthesis and storage of food materials.

Explain the three different types of plastids and their corresponding functions. (3 marks)